# Project 2 Report: Fitbit Fitness Tracker Data

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## Summary


- **Outliers (Section 1):** the plots had a few clear outliers, such as the daily heart rate spikes of one user (up to about 160 beats per minute), 14 days with 0 steps for one user (13 of them with 1,440 sedentary minutes, which looks like the device not being worn), and a jump of 1.2 kg in one day in a manually entered weight record.
- **Calories (3.1):** calories per hour are more strongly correlated with intensity ($r = 0.897$) than with steps ($r = 0.815$).
- **Sleep (3.2 and 3.4):** minutes asleep and time in bed are almost on a line ($r = 0.93$), while minutes asleep and sedentary minutes have a negative correlation ($r = -0.60$). This is mostly because of how the minutes of the day are divided up.
- **Users (3.3):** three randomly chosen users differ in when their intensity peaks (8:00, 9:00 and 17:00) and in how high the peaks are.
- **T-tests (Section 4):** day vs. night had the most significant difference in calories per hour (about 34.5 calories, $p = 1.7 \times 10^{-10}$), while odd vs. even days of the month had the least (about 0.6 calories, $p = 0.42$).


## Introduction

This report explores the Fitbit Fitness Tracker dataset from Kaggle, which has data from 33 users over the 31 days from April 12 to May 12, 2016. The data comes from 18 CSV files with daily, hourly and per-minute activity, sleep, heart rate and weight records. Not every user has data in every category.

Section 1 looks at the distributions of heart rate, sleep, steps and weight, with a focus on outliers. Section 2 describes data cleaning and merging. Section 3 looks at the correlations between pairs of variables and the differences between users. Section 4 uses t-tests to find the groupings with the most and least significant differences in calories burned, and Section 5 is my reflections on the work.

# 1 Distributions and Outliers

## 1(a) Heart Rate

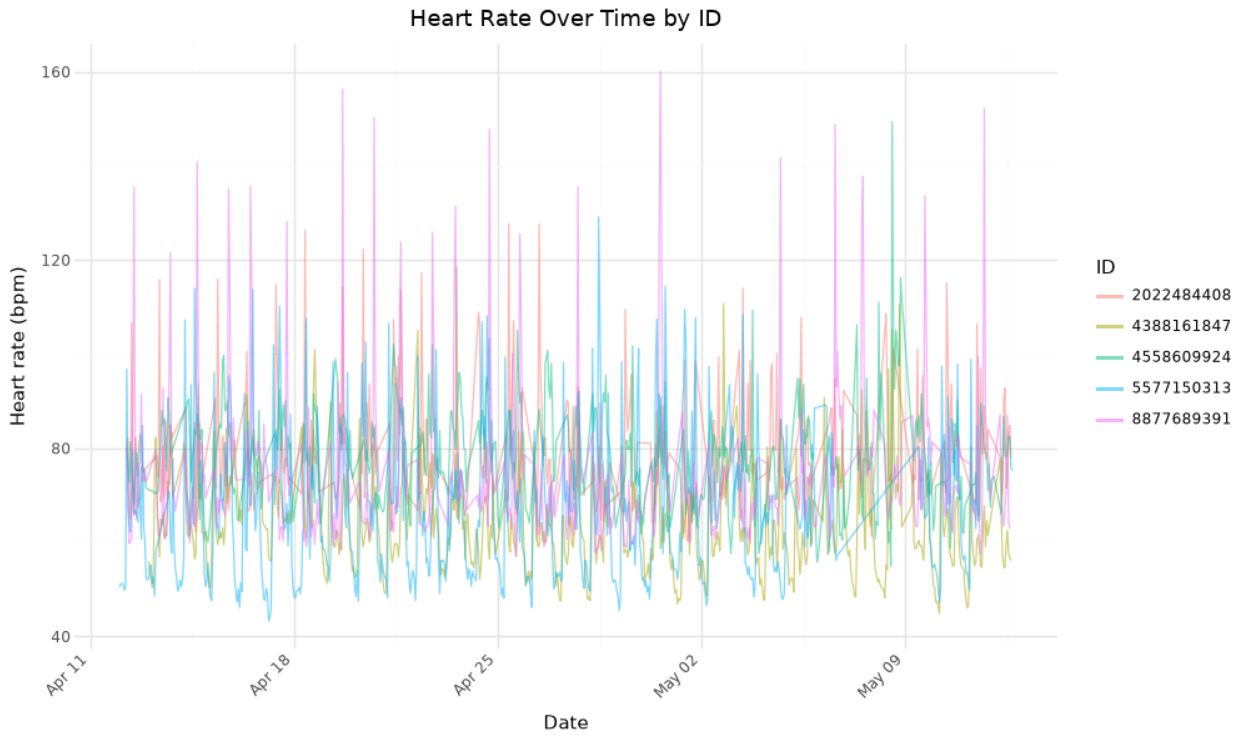



Figure 1: Hourly mean heart rate over time for five users.

Out of the 14 users with heart rate data, 7 of them have at least 28 days of readings, and I picked 5 of them randomly using `random_state=1`. The readings were about every 5 seconds, so I decided to take the mean of each minute and then the mean of each hour to make plotting them on one date axis readable. The main tradeoff of this was that an hourly mean hides short peaks, so the real peaks are probably higher than what the plot actually shows. Most of the values for all 5 users fall between 45 and 120 beats per minute, and there seems to be a similar rise and fall daily for everyone.

User 8877689391 has sharp spikes that reach about 115 to 160 beats per minute, roughly once a day. An hourly mean that high suggests that the user might be giving a sustained effort at something like a workout, but of course this can't be verified. User 4558609924 has a spike of about 150 near May 8th, and 5577150313 has a few spikes of 110 to 130. The lowest values are around 43 to 50, and belong to users 5577150313 and 4388161847, which could be their resting heart rates during sleep. There's also a straight line from May 6 to May 9 for user 5577150313, which is just the plot connecting two points that have a gap in the data.

## 1(b) Minutes Asleep

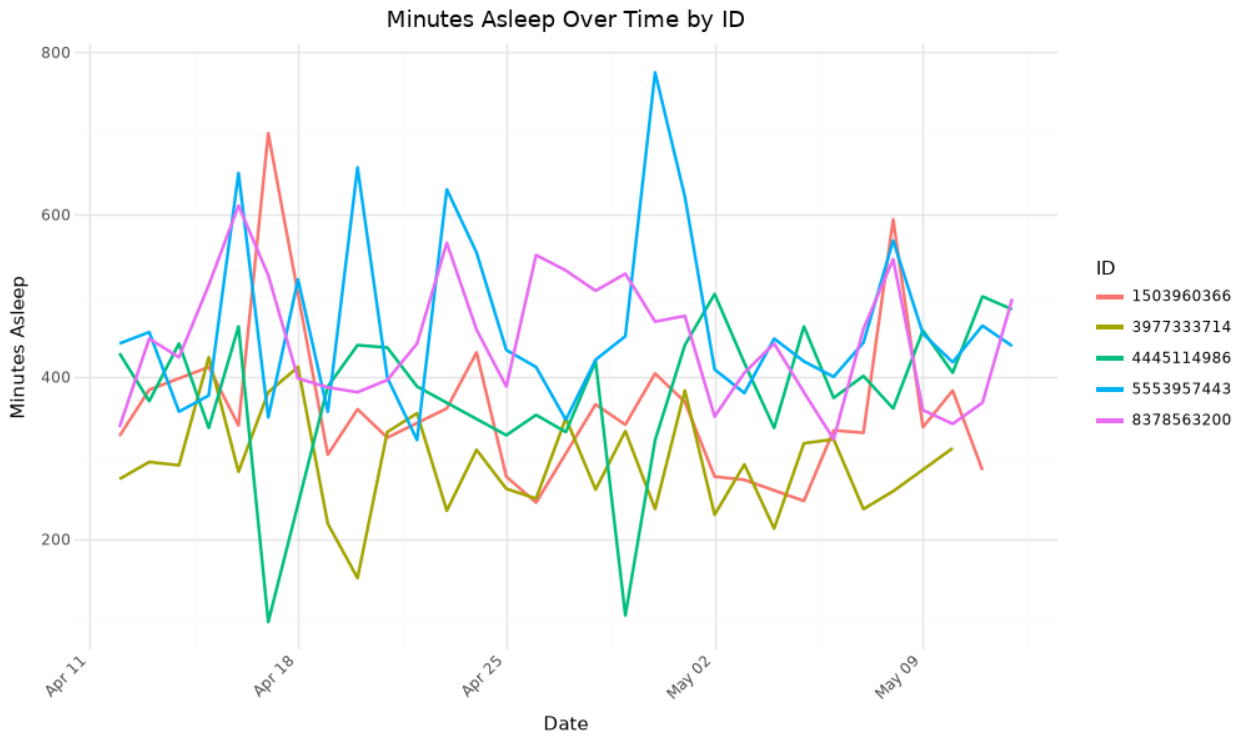



Figure 2: Minutes asleep over time for five users.

I used `TotalMinutesAsleep` from `sleepDay`, picking 5 out of the 10 users with at least 25 nights at random using `random_state=1`. Missing data is just filled in with a line between the two surrounding nights, and 3 of the users are missing at least 3 of the 31 days: 1503960366 is missing 6, and 3977333714 and 4445114986 are missing 3 each.

There are a few outliers in the data. User 5553957443 has 775 minutes asleep on April 30th, which is the highest value out of the 5 users, and on April 17th 1503960366 had a night of about 700 minutes. On the other end of the scale, user 4445114986 has two shorter nights of 98 and 106 minutes on April 17th and 29th respectively. All of these outliers look odd because they're far from the user's usual sleep pattern, and I think the short ones might have been days where the tracker was only worn for part of the night, or where only a nap was recorded. The longer ones could be real data, with the user just sleeping for a long time, but I have no way of verifying the reason for the outliers.

## 1(c) Total Steps

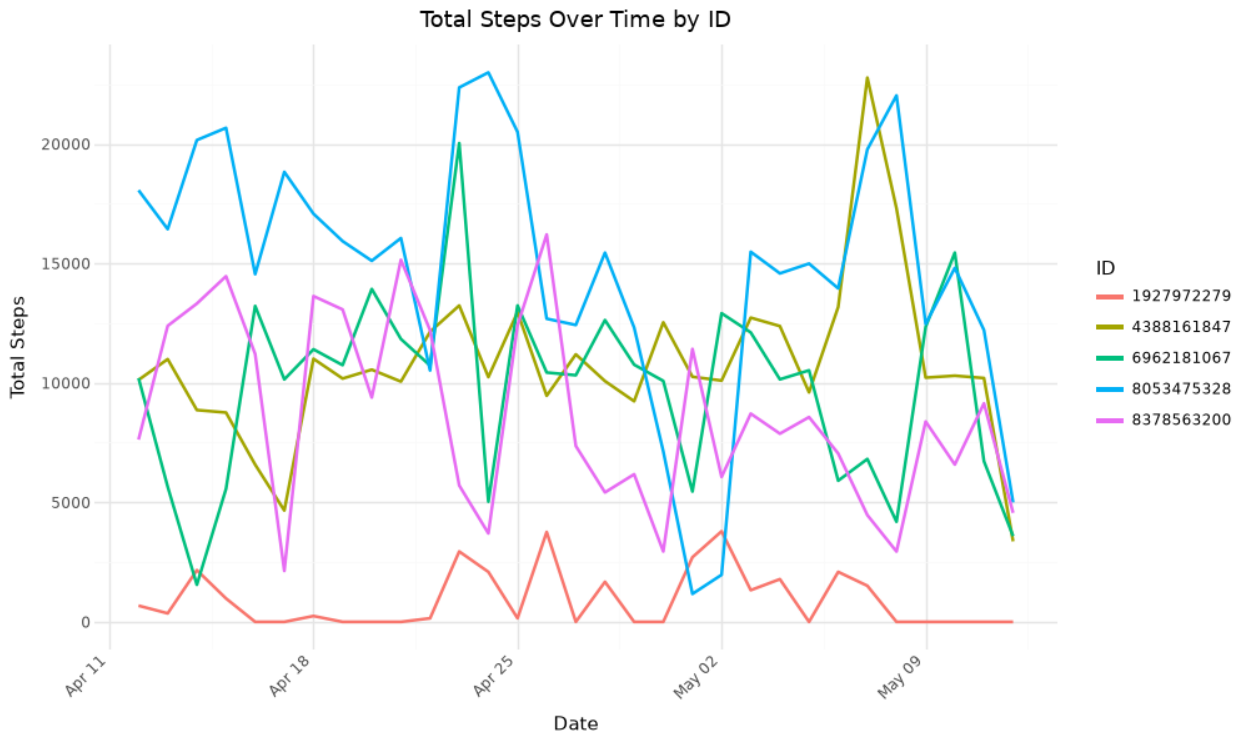



Figure 3: Total steps per day over time for five users.

For this plot, I used `TotalSteps` from the `dailyActivity` dataset, picking 5 of the 21 users who have all 31 days at random. Four of the users tend to have between 5,000 and 15,000 steps per day, with means of about 8,700 to 14,800.

The biggest outlier in this data is user 1927972279, who has at most 3,790 steps per day, and 14 days with 0 steps. On 13 of those 14 days, the recorded sedentary minutes were 1,440 (a full day), so I think these were days when the device wasn't worn or carried, rather than days with no walking at all, but that's just a guess. These days also pull down any averages that include this user. There are also high outliers, with user 8053475328 having a day of about 23,000 steps on April 24th and user 4388161847 having a day of about 22,800 steps on May 7th, which is high but still possible. For the other users, 8053475328 had a low of about 1,200 steps and 6962181067 had a low of about 1,600 steps. These days were much lower than their usual, and could be attributed to rest days or the device not being worn for part of the day.

### 1(d) Weight

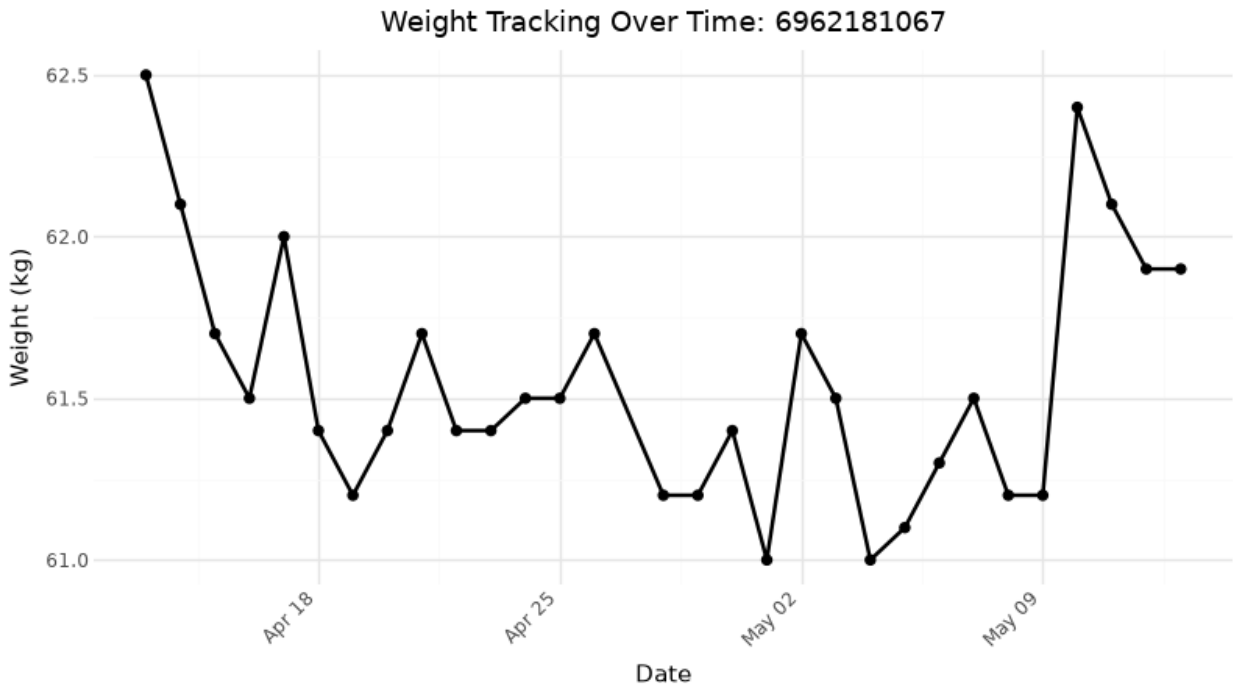



Figure 4: Weight over time for the user with the most records.

This plot is only for user 6962181067, whose weight records cover 30 out of the 31 days, the most of any user. All 30 records are manual entries (`IsManualReport` is true for every one), so they were entered by hand rather than coming from a smart scale, and the weights are rounded to 0.1 kg. The user's weight stays between 61.0 and 62.5 kg for the whole month, starting at 62.5 and ending at 61.9.

The y-axis is only between 61 and 62.5 kg, so the changes look much bigger than they actually are. In the first few days, there's a quick 1 kg drop from 62.5 to 61.5 kg, which stands out, and the 1.2 kg jump from 61.2 to 62.4 kg on May 9th does as well. Someone's body weight changing by over 1 kg in a day or two is more than what I'd expect normally, and it could be affected by the time of day the measurement was taken, how much food or water was in the body, or a miscalibrated scale. There also seems to be a straight line between April 25 and April 27, because there's no record for April 26.

## 2 Data Processing

### 2.1 Hourly and Minutely Data

To build the hourly dataframe, I merged `hourlySteps`, `hourlyIntensities`, and `hourlyCalories` on the user `Id` and the datetime of each reading. Similarly, the minutely dataframe merges `minuteCaloriesNarrow`, `minuteIntensitiesNarrow`, and `minuteMETsNarrow`. I used outer merges and also checked that each merged dataframe was one-to-one, so that a duplicate key would raise an error. In terms of shape, the hourly dataframe has 22,099 rows and 33 users, and the minutely dataframe has 1,325,580 rows and the same set of 33 users. Neither dataframe has missing values.

The time columns were stored as 12-hour strings like `4/12/2016 12:00:00 AM`, so I converted them with an explicit format and put them into a new column named `Date`. I did the same for `dailyActivity` and `sleepDay`. I checked that the conversion worked by confirming that the dates only fall in April and May 2016, that all 24 hours appear, and that 12:00 AM and 12:00 PM were read correctly. After creating `Date`, I chose to drop each dataframe's respective date column (`ActivityHour`, `ActivityDate`, `SleepDay`, etc.).

```
Hourly (only 00:00 and 12:00 rows):
            Id  Calories                Date  TotalIntensity  AverageIntensity  StepTotal
0   1503960366        81 2016-04-12 00:00:00              20          0.333333        373
12  1503960366        73 2016-04-12 12:00:00              11          0.183333        253
24  1503960366        69 2016-04-13 00:00:00              14          0.233333        144
36  1503960366        82 2016-04-13 12:00:00              15          0.250000        486

Minutely:
           Id  Calories                Date  Intensity  METs
0  1503960366    0.7865 2016-04-12 00:00:00          0    10
1  1503960366    0.7865 2016-04-12 00:01:00          0    10
2  1503960366    0.7865 2016-04-12 00:02:00          0    10
3  1503960366    0.7865 2016-04-12 00:03:00          0    10

dailyActivity (selected columns):
           Id  TotalSteps  LightlyActiveMinutes  SedentaryMinutes  Calories       Date
0  1503960366       13162                   328               728      1985 2016-04-12
1  1503960366       10735                   217               776      1797 2016-04-13
2  1503960366       10460                   181              1218      1776 2016-04-14
3  1503960366        9762                   209               726      1745 2016-04-15

sleepDay (413 rows, before removing duplicates):
           Id  TotalSleepRecords  TotalMinutesAsleep  TotalTimeInBed       Date
0  1503960366                  1                 327             346 2016-04-12
1  1503960366                  2                 384             407 2016-04-13
2  1503960366                  1                 412             442 2016-04-15
3  1503960366                  2                 340             367 2016-04-16
```

`sleepDay` contains three exact duplicate rows, which I removed in Section 1(b) and wherever else I used it, so that those days weren't counted twice.

## 2.2 Heart Rate

The heart rate file has a new reading about every five seconds, so I rounded each timestamp down to the minute and took the mean of each minute's readings to get the average heart rate per minute. This left me with 333,420 rows for 14 users. I stored the result in its own dataframe instead of adding it to the minutely dataframe, because only 14 out of the 33 users have heart rate data, and the heart rate data only runs until 16:20 on May 12th, while the minutely data ends at 15:59. Storing heart rate data in its own dataframe avoided dropping heart rate minutes after 15:59 or adding missing values to other columns.

```
             Id                Date      Value
275  2022484408 2016-04-12 12:00:00  69.272727
276  2022484408 2016-04-12 12:01:00  59.375000
277  2022484408 2016-04-12 12:02:00  59.333333
278  2022484408 2016-04-12 12:03:00  62.166667
```

# 3 Correlation and Plots

## 3.1 Calories vs. Steps and Intensity




Figure 5: Calories per hour vs. steps per hour.




Figure 6: Calories per hour vs. total intensity per hour.

Both plots use the data in the hourly dataframe, which has 22,099 rows and one point for each user and hour. The transparency of the plotted points was set to 0.15 so that dense areas can be told apart from single points. For the measure of intensity I used `TotalIntensity`, because `AverageIntensity` is just that divided by 60, and it gives the same correlation (see Section 5). Calories had a correlation of $r = 0.815$ with steps and $r = 0.897$ with intensity, which means intensity is the stronger linear predictor. Steps and intensity are also strongly correlated with each other, with $r = 0.896$, so they carry a lot of the same information.

For both plots, the points don't go through zero. Even at 0 steps or 0 intensity, there is a baseline rate of about 50 to 100 calories, which I think is the calories the body burns just to stay alive. The steps plot is also more spread out at any given step count. For example, calories range from about 125 to over 400 at the 2,500 steps per hour mark, and there also seem to be two separate groups of points. My guess is that steps only count when walking or running, while intensity also measures effort in activities that don't add many steps, like cycling or lifting weights, so intensity separates those hours better. However, I can't confirm that theory just based off this data.

There are 16 rows that have 0 steps and over 600 calories, most of which show up as the dark point at 0 steps and about 670 calories on the steps plot. I explain the reasoning behind this point in Section 5. I left those outliers in the plots, but removing them gave correlations of $r = 0.846$ for steps and $r = 0.893$ for intensity. My answer doesn't depend on those 16 rows, as intensity still has the higher correlation, but the gap did shrink from 0.082 to 0.048. Based on this, it's reasonable to conclude that intensity is more relevant to calories than steps, although correlation alone doesn't show that intensity causes the calories burned, and both are measured on the same device.

### 3.2 Minutes Asleep vs. Time in Bed

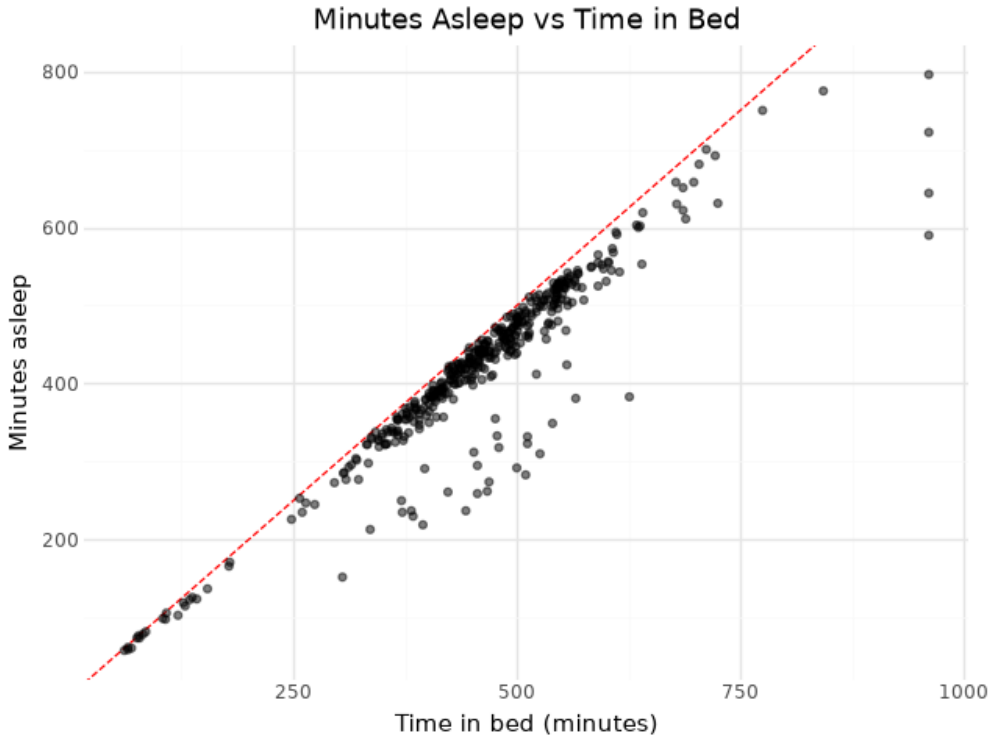



Figure 7: Minutes asleep vs. time in bed (dashed line: $y = x$).

This plot uses the `sleepDay` dataframe, which has 410 nights of data over 24 users after removing 3 duplicate rows. The points follow the dashed $y = x$ line very closely, and the correlation between the two columns is $r = 0.93$. None of the nights had more minutes asleep than minutes in bed, so every point is on or below the line. This is what I'd expect from the data because time spent asleep is part of the time in bed.

The distance below the line is the time spent in bed, but not asleep. The median gap was 25.5 minutes, but there was a lot of variation, with a maximum of 371 minutes. Out of the 410 nights, 52 of them (about 13%) had a gap of over an hour, with 30 of them having a gap of over two hours. Shorter nights are closer to the line, and most of the points further below the line are at a range of 300 to 600 minutes spent in bed.

Nights with larger gaps aren't spread evenly among the users. User 3977333714 accounts for 28 of the 52 nights with a gap of over an hour, and all of that user's recorded nights had a gap over an hour. This could mean that they regularly spend a lot of their time in bed awake. It could also mean that their tracker records sleep differently, but I can't know just based off the data.

There are also four nights with the maximum time spent in bed over the whole dataset, at 961 minutes, and they account for the column of points at the right side of the plot. Three are from a single user and one is from a different user, and their minutes asleep range from about 590 to 800. Although I can't confirm it, I suspect that, rather than four nights being the exact same length, there may be a limit in how the tracker records time in bed.

Overall, minutes asleep is a very good predictor of time in bed, but time in bed is never smaller than minutes asleep, and the size of the gap between the two depends on the user and the night.

### 3.3 Intensity over the Day

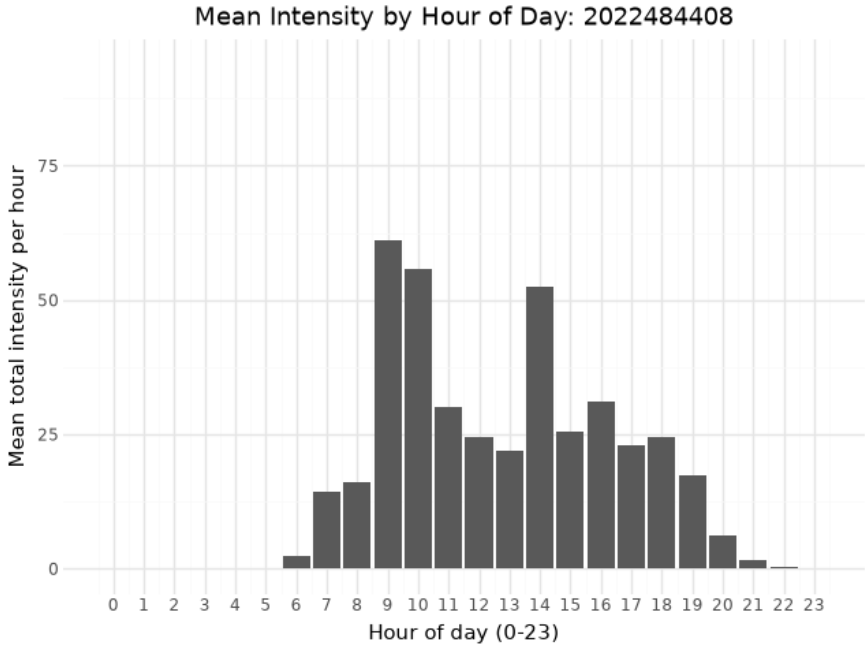



Figure 8: Mean total intensity by hour of day, user 2022484408.

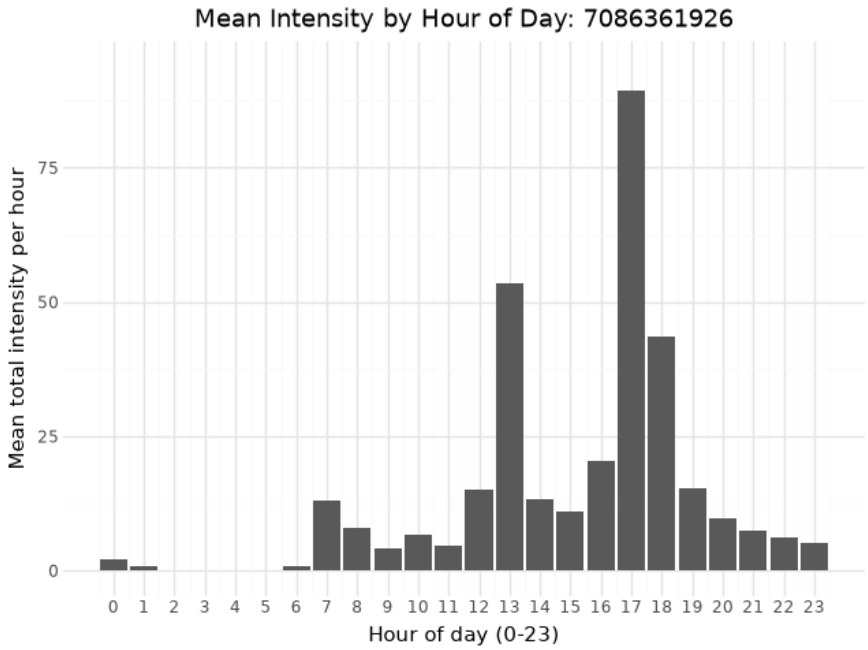



Figure 9: Mean total intensity by hour of day, user 7086361926.

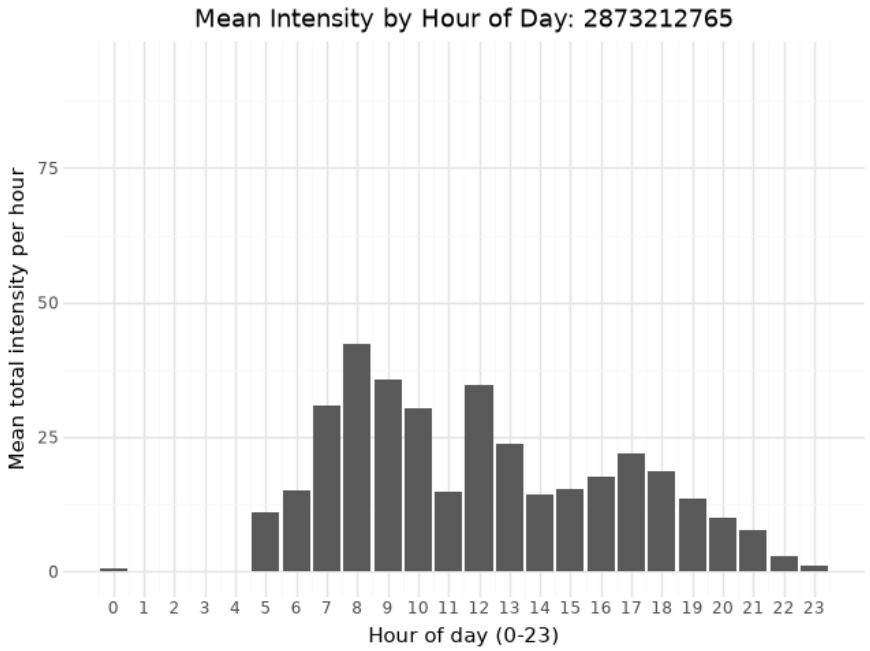



Figure 10: Mean total intensity by hour of day, user 2873212765.

Out of the 19 users with all 31 days of data, I picked 3 users at random (`seed=1`), so that each bar is an average over roughly the same number of days. Each bar is the mean `TotalIntensity` for the given hour of the day, and the three plots share the same y-axis so the heights can be compared directly. Because the bars represent means, they don't show how much an hour varies from day to day.

All 3 of the users have almost no intensity from about 1:00 to 5:00, which is around the time I'd expect people to be asleep, but after that intensity tends to differ a lot. User 2022484408 is active from about 6:00 to 21:00, with the highest bar being at 9:00 (about 61), and with a second peak at 14:00 (about 52). User 7086361926 has a much lower baseline throughout the day (mostly under 20, outside the afternoon), but has two sharp spikes at 13:00 (about 53) and 17:00 (about 89). User 2873212765 has the flattest pattern starting at 5:00, peaking at 8:00 (about 42), and then staying between about 15 and 35.

The users differ in both timing and peak height. User 7086361926 peaks in the late afternoon at 17:00, while the other two users peak in the morning, at 9:00 and 8:00. The peak of 7086361926 is also about twice as tall as the peak of 2873212765, and about 1.5 times as tall as the peak of user 2022484408. The users also end the day at different times, with 2022484408 being close to zero at 22:00, while 7086361926 is close to zero at 22:00 and at 0:00 to 1:00.

To check whether a tall bar comes from a regular habit or if it comes from a few extreme days, I also looked at the daily values at each user's peak hour. For 7086361926 at the peak hour of 17:00, the median was 126.5, and 19 out of the 31 days were above 60, so the peak looks like it's a regular habit. User 2022484408 had a more irregular peak, with a median of only 35 at 9:00 with a standard deviation of 53, and 13 of the 31 days were above 60. For 2873212765 at 8:00 the median was 33 and only 5 of 31 days were above 60, so the peak is pretty modest and the height is normal for that hour. However, the data can't tell me what the users were actually doing during the day, for example whether the spike at 17:00 is an exercise routine or something else.

### 3.4 Minutes Asleep vs. Sedentary Minutes

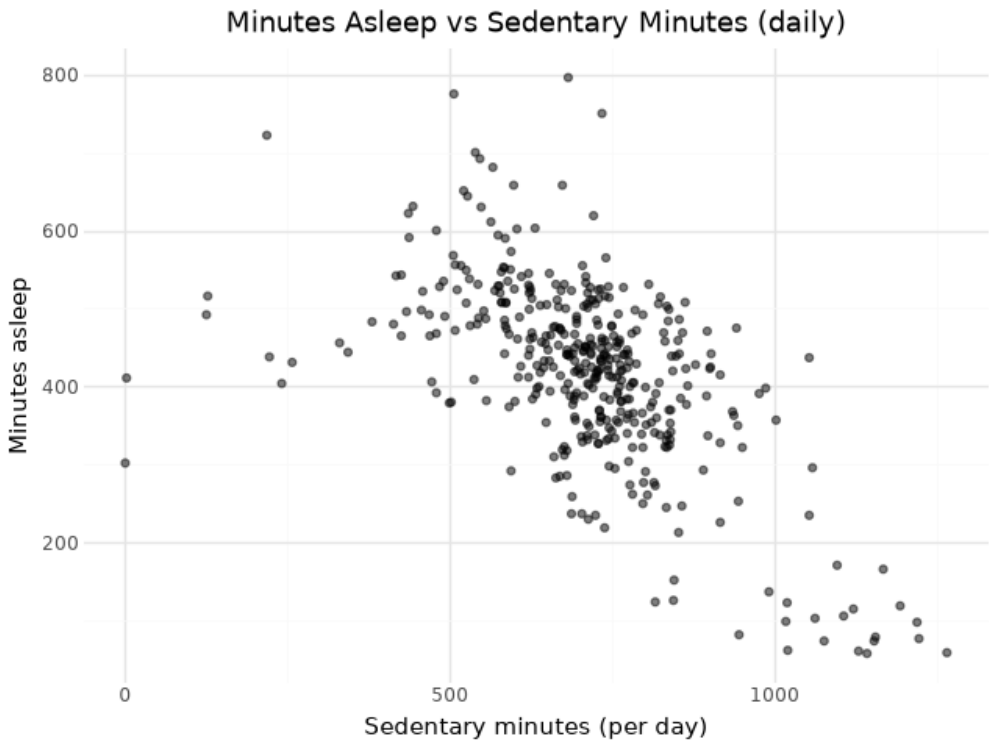



Figure 11: Minutes asleep vs. sedentary minutes per day.

This plot uses the 410 nights in `sleepDay`, after removing the 3 duplicates, joined to `dailyActivity` on the `Id` and `Date` columns. All 410 nights had a matching activity row, covering 24 users. The relationship is negative and moderately strong, with $r = -0.60$, meaning that days with more sedentary minutes tend to have fewer minutes asleep. There are 21 days with over 1000 minutes, and all of them have between 58 and 437 minutes asleep.

This result was the opposite of what I expected, because I had assumed that time asleep would also be counted as sedentary time, making their relationship positive. To test that, I added up the four activity-minute columns (sedentary, lightly active, fairly active, and very active) for each of the 410 nights. I found that none of them added up to 1440 minutes, which means that some of the minutes in the day weren't in any of these four columns. The missing minutes have a median of 457, which is close to the median of 432.5 minutes asleep, and they correlate with minutes asleep at $r = 0.63$. This suggests that sedentary minutes mostly leave out the time spent asleep. If that's the case, the negative correlation is largely a matter of how the day is divided up, as more sleep means there is less time for a person to be sedentary. However, this explanation isn't proven, because the data doesn't say how the tracker assigns minutes to categories.

There are some other possible causes that could be partly behind the pattern. The relationship isn't only a difference between users, as 13 out of the 15 users with at least 10 nights had a negative correlation within their own nights (from $-0.23$ to $-0.84$), and the other two were close to zero. A behavioral cause of this could be that people sleep less on days that they're very inactive, or that sleeping for very little time could lead to a more sedentary day, but this can't be shown from the data alone. A day spent working at a desk or being sick could also be a factor that affects both numbers. Because of these other potential factors that aren't measured, correlation alone can't

show that sedentary time causes less sleep, or vice-versa.

There are also a few unusual points in the data. Two of the nights have about 0 sedentary minutes, along with very few steps on the same day, and a few others have under about 300. These days could be times when the tracker wasn't worn for most of the day, but I can't confirm that. I left them in the plot, as there aren't very many of them and the trend is still clear.

# 4 T-tests

## 4.1 Setup and Predictions

For both tests, the outcome was `Calories` per hour from the hourly dataframe. I used a significance level of $\alpha = 0.05$ with two-sided tests. My predictions before running anything were that day versus night would show the biggest difference in calories, because people are usually asleep at night and will only burn some baseline amount of calories. For my case, I defined night as being from 22:00 to 05:59, with day being 06:00 to 21:59. These boundaries are mostly subjective, and are mostly based off my own sense of when people sleep, but also keeping in mind the intensity by hour plots in Section 3.3, where intensity was close to zero from about 1:00 to 5:00. I predicted that there would be the smallest difference between an odd versus an even day of the month, because it wouldn't make sense for what day it is to change how many calories someone burns, which makes it a good control group.

## 4.2 Choice of Test

Rows that come from the same user, and especially consecutive hours from the same user, are not independent of each other. A two-sample t-test on all 22,099 hourly rows would treat them as independent, which might make the results look more concrete than they actually are. To avoid that, I started by averaging each user's calories per hour within each group, and used one pair of values for each of the 33 users to run a paired t-test (`scipy.stats.ttest_rel`) on them. Then, each user is compared against themselves, which also removes variables such as body size differing between users from the comparison. The biggest tradeoff of this methodology is that the sample size drops from 22,099 rows to just 33 pairs.

For both comparisons, the null hypothesis is that the mean of the per-user differences is zero, or $H_0 : \mu_d = 0$, while the alternative is that it's not equal to zero, $H_1 : \mu_d \neq 0$. For the first comparison, $d$ is the user's mean calories per hour during the day minus the mean at night. And for the second, $d$ is the user's mean on odd days minus that on even days.

## 4.3 Results

All 33 users had data in every group. Table 1 shows the results, and the effect size is Cohen's $d_z$, the mean difference divided by the standard deviation of the differences.

| Comparison | Mean difference | 95% CI | $t(32)$ | $p$ | $d_z$ |
|---|---|---|---|---|---|
| Day − night | 34.48 | (26.85, 42.11) | 9.21 | $1.7 \times 10^{-10}$ | 1.60 |
| Odd − even day | 0.58 | (−0.87, 2.03) | 0.81 | 0.42 | 0.14 |

Table 1: Paired t-tests on per-user mean calories per hour (33 users). Differences are in calories per hour.

Users burned an average of 108.25 calories per hour during the day, which was 34.48 more than the 73.77 calories per hour burned at night. This is a large effect ($d_z = 1.60$), and I rejected the null hypothesis ($t(32) = 9.21$, $p = 1.7 \times 10^{-10}$). However, for odd versus even days, the averages were 97.00 and 96.42 calories per hour, which is a difference of just 0.58. For this comparison, I failed to reject the null hypothesis ($t(32) = 0.81$, $p = 0.42$). The confidence interval for this difference includes zero, so I found no evidence that calories differ depending on the day, but the interval can't rule out any difference of up to about 2 calories per hour.

### 4.4 Comparison with Welch's Test

For further comparison, I ran Welch's two-sample t-test on all hourly rows, without averaging by user. For day versus night, it found a difference of 34.80 calories per hour (95% CI 33.42 to 36.18), which was almost the same as the paired test. The main difference was that this had a much more extreme test statistic ($t = 49.5$, $p$ pretty much zero). For odd versus even days, it found a difference of 0.22 calories per hour ($t = 0.27$, $p = 0.79$). The conclusions based on Welch's tests are the same as the paired tests, but the p-value for day versus night is much smaller due to the large number of rows. The rows are also not independent, so I treat the paired test as the main result.

### 4.5 Interpretation and Limitations

Based on the data, both of my predictions were supported. The large difference in day versus night is plausible because most hours during the night are spent sleeping, when calories burned are mostly baseline, and the odd or even comparison acted as a control group with no meaningful difference. There were a few limitations with my testing though, as the paired t-test assumes that the per-user differences are roughly normal, which I did not check with only 33 users. Each user counts equally to the comparison regardless how many days of data they have, and one user only had 88 hourly rows. I also only ran tests with two groupings, so while the results confirm my predictions for those two, I didn't search over all possible groupings to show that they really are the largest and smallest differences.

## Conclusion

In this report, I explored the Fitbit dataset by looking at outliers, merging the files into hourly and per-minute dataframes, comparing pairs of variables, and testing for differences in calories burned. Intensity was a better linear predictor of calories per hour than steps, minutes asleep were closely tied to time in bed, and the most significant difference in calories per hour was between day and night, while the least significant was between odd and even days of the month. Several of the results turned out to depend on how the data was recorded, such as the 1,440 sedentary minutes on days when a device was probably not worn, and the fact that it seems like time spent asleep is excluded from sedentary minutes.

There are also some limitations to the data, and therefore what the results show. It's a small dataset, with just 33 users over 31 days, and I used samples of 3 to 5 users in some of the plots, so the differences between users may not describe other people. The correlations also don't show what causes what, and my explanations for the outliers are just guesses, as I don't have any way of determining why a value is unusual just by looking at the data. If I were to continue working on this, I would definitely check the causes I suggested. For example, I would look at the minute-level data around the outliers, then test more groupings for the t-tests instead of only two.

# 5 Reflection

## 5.1 Something I Understand Differently Now

While working on 3.4, I expected `SedentaryMinutes` to include the time recorded in `TotalMinutesAsleep`. That would have meant they would have a positive relationship, where more sleep would go with more sedentary minutes. Instead, the two had a correlation of $r = -0.60$. To figure out why, I checked whether the sedentary, lightly active, fairly active, and very active minutes added up to the 1,440 minutes in a day on days with a sleep record, and it didn't add up once out of 410 days. The leftover minutes, which was calculated as 1,440 minus the sum of the four columns, had a median of 457, which was close to the median of 432.5 minutes asleep, and they had a correlation with minutes asleep at $r = 0.63$. This suggests that the tracker mostly leaves sleep out of sedentary time. A day always has 1,440 minutes, so more sleep leaves fewer minutes awake that can count as sedentary, and the negative sign is largely built into how the minutes are counted, rather than showing that sitting more negatively affects sleep. I can't confirm the tracker's exact rule based on the data, but now I understand that what a column measures depends on how the device defines it, not only what it's called.

## 5.2 A Decision I Made

When working with intensity in 3.1, I wasn't sure whether to use `TotalIntensity` or `AverageIntensity` from the hourly data. I thought that they might be the same measurement but on different scales, so I checked the data. `np.allclose` comparing `AverageIntensity` with `TotalIntensity / 60` returned False, but the largest difference between them was only about $3.3 \times 10^{-7}$, which is something I'd expect if the values of `AverageIntensity` are rounded to six decimals. So the False most likely came from rounding and not from a real difference. The check returned True when using a tolerance of $10^{-6}$. Both columns also had the same correlation with calories ($r = 0.8966$) and gave the same scatterplot shape. Because they gave identical results, I chose to use `TotalIntensity` because it's an hourly total like steps and calories, which makes the plots easier to compare.

## 5.3 A Moment of Uncertainty

When plotting calories against steps and against intensity, I noticed that there was one especially dark point at 0 steps and about 670 calories in the steps plot, and it appeared at an intensity of around 165 out of 180 in the intensity plot. My first interpretation was that it might be exercise that just doesn't register as steps, such as cycling or rowing. But the point was solid black even though I had set the transparency to 0.15, so it couldn't just be a single observation. I filtered for rows with 0 steps and over 600 calories, and I found that there were 16 rows that were all from one user, and all in the 05:00 hour on different days, with 15 of them having exactly the same values (669 calories and 165 intensity). It would be very unlikely for real exercise to produce exactly identical values on 15 different mornings, so I think it's more likely that there was a recording error or some other artifact, but I can't confirm the cause just from looking at the data. By removing the 16 rows, the correlation for steps changed from 0.815 to 0.846, and from 0.897 to 0.893 for intensity, so my conclusion that intensity is the better predictor actually didn't depend on them.

### 5.4 Looking Back

If I did this again, I would start by checking every file for repeated or duplicate rows and suspicious maximum values, before interpreting any outliers. The 16 identical rows in 3.1 and the four nights with exactly 961 minutes in bed, which was a maximum in the dataset and from 2 different users, both looked like interesting findings when I first noticed them. I spent a long time trying to explain them before deciding to examine the data behind them. When inspecting the sleep data, I did find three exact duplicate rows, which kept me from double counting them. Next time, I would just write a short script that prints the most common repeated rows and each column's maximum for every file before making any plots.